KBTU · MACHINE LEARNING · LAB 2

Life Satisfaction Prediction using GDP per Capita

Report on the laboratory work: linear regression and k-nearest neighbors regression in scikit-learn

1. Goal and objectives

The goal of the work is to predict how satisfied people in a country are with their lives using a single feature: GDP per capita. The work reinforces three skills from the lectures:

- loading and visualizing real-world data with `pandas` and `matplotlib`;
- training and comparing two regression models with scikit-learn: linear regression and k-nearest neighbors (KNN);
- interpreting model outputs and discussing how well the models generalize to new data.

2. Dataset

The file `lifesat.csv` contains 27 countries and three columns: `Country`, `GDP per capita (USD)` and `Life satisfaction` (a score from 0 to 10). `df.info()` shows there are no missing values: all 27 rows are non-null in every column. Summary statistics from `df.describe()`:

Statistic	GDP per capita (USD)	Life satisfaction
count	27	27
mean	41,565	6.57
std	9,631	0.77
min	26,456	5.40
25%	33,938	5.90
50%	41,627	6.80
75%	49,691	7.30
max	60,236	7.60

All countries have a GDP per capita between about 26,000 and 60,000 USD, and life satisfaction ranges from 5.4 to 7.6. This narrow range is important later: the models have never seen poorer or richer countries.

3. Data exploration

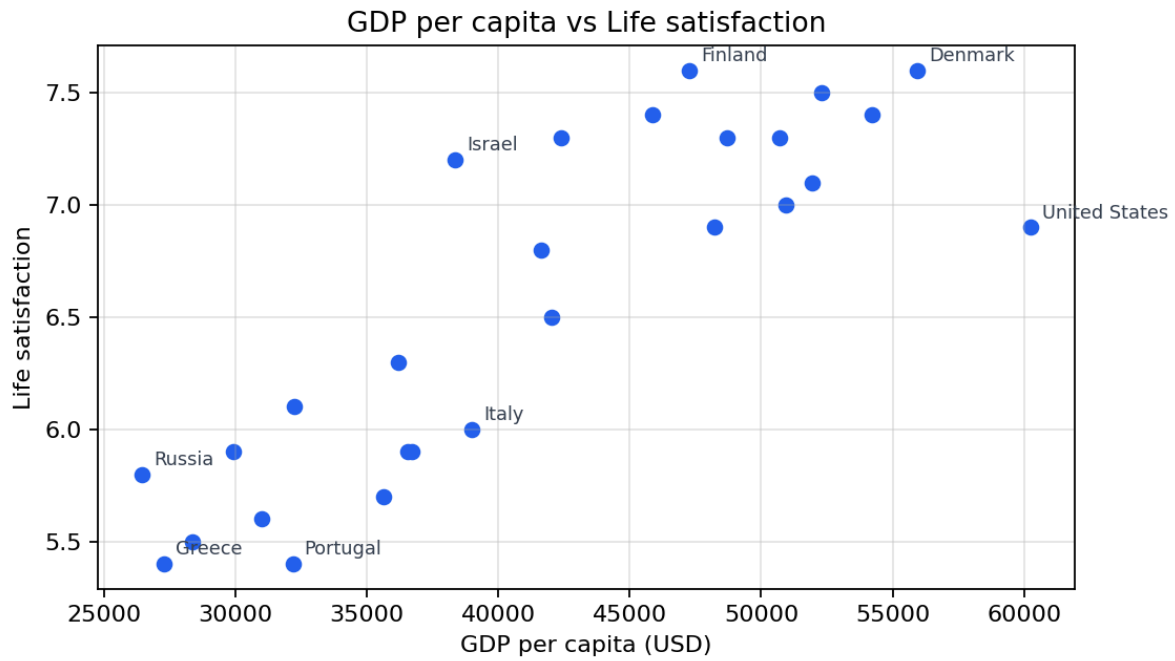


Figure 1. GDP per capita vs life satisfaction (Q3).

The scatter plot shows a **clear positive, roughly linear relationship**: richer countries tend to report higher life satisfaction. The Pearson correlation is **r ≈ 0.85**, which is a strong correlation. The points do not lie on a perfect line, though. Israel reports 7.2 at a moderate GDP, while the United States, the richest country in the sample, reports only 6.9. So GDP explains a large part of the differences, but other factors matter too, and correlation alone does not prove that money causes happiness.

4. Linear regression

4.1 Model

The input matrix and the target were extracted as `X = df[["GDP per capita (USD)"]].values` and `y = df[["Life satisfaction"]].values`. Both have shape **(27, 1)**. Double brackets keep the data two-dimensional, because scikit-learn expects the features as a matrix of shape (n_samples, n_features).

Linear regression models the target as a straight line:

$$\hat{y} = \theta_0 + \theta_1 \cdot x$$

The parameters are found by the least squares method, which minimizes the mean squared error MSE = $(1/m)\ \Sigma\ (\hat{y}_i - y_i)^2$.

4.2 Results

Quantity	Value	Meaning
Coefficient θ_1 (`coef_`)	6.779e-05	+1,000 USD of GDP ≈ +0.068 points of satisfaction
Intercept θ_0 (`intercept_`)	3.749	prediction at GDP = 0 (an extrapolation with no practical meaning)
R^2 on training data	0.73	the line explains ~73% of the variance
RMSE / MAE on training data	0.39 / 0.31	typical error in satisfaction points

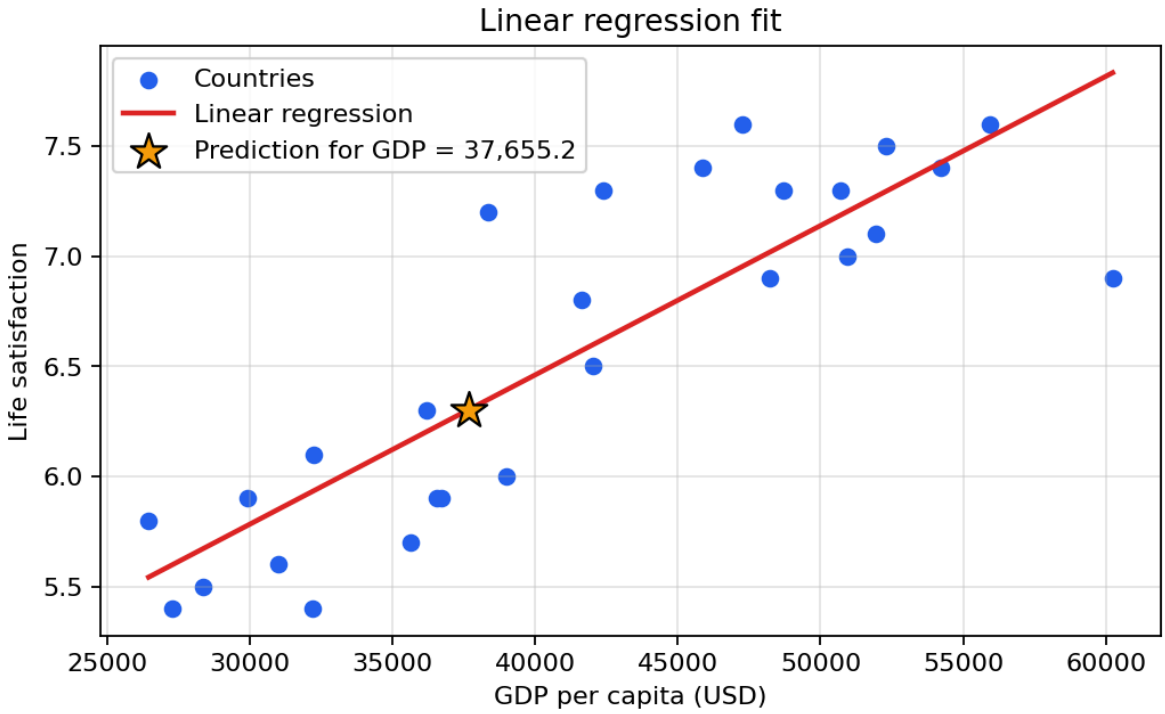


Figure 2. Regression line over the data (Q6) and the prediction for GDP = 37,655.2 USD (Q7).

4.3 Prediction for GDP = 37,655.2 USD

$$\hat{y} = 3.749 + 6.779\text{e-}05 \times 37{,}655.2 \approx \mathbf{6.30}$$

The predicted life satisfaction is about **6.30**. The value is plausible: it lies between the scores of countries with similar GDP (Lithuania and Slovenia 5.9, Italy 6.0, Spain 6.3, Israel 7.2) and follows the overall upward trend.

5. K-nearest neighbors regression

5.1 Model

KNN regression does not learn parameters. It stores the training data, and for a new point it finds the *k* closest training points (by Euclidean distance) and returns the **average** of their target values. This is *instance-based* learning, as opposed to the *model-based* linear regression.

The five countries closest to GDP = 37,655.2 USD are:

#	Country	GDP (USD)	Distance	Life satisfaction
1	Israel	38,341	686	7.2
2	Lithuania	36,732	923	5.9
3	Slovenia	36,548	1,107	5.9
4	Italy	38,992	1,337	6.0
5	Spain	36,215	1,440	6.3

For k = 3 the prediction is (7.2 + 5.9 + 5.9) / 3 ≈ **6.33**. It differs from the linear regression prediction (6.30) by only 0.032: both models agree because the new point lies inside the range of the training data.

5.2 Effect of the number of neighbors

k	Prediction for GDP = 37,655.2	R^2 on training data
1	7.20	1.00
3	6.33	0.85
5	6.26	0.81
10	6.37	0.82

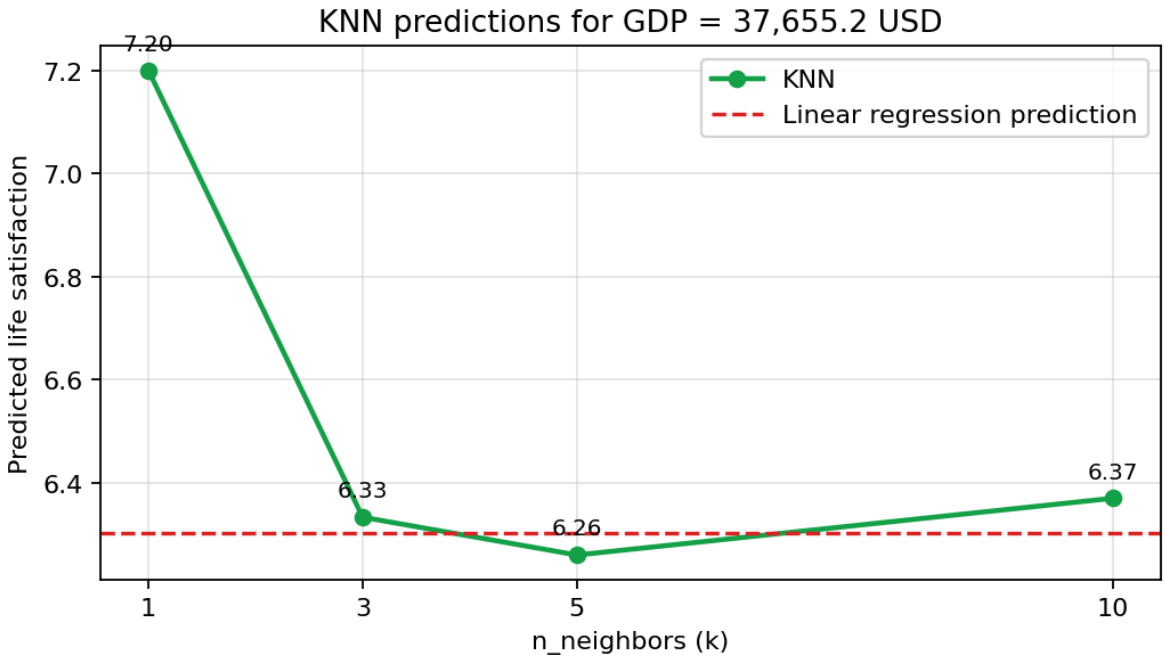


Figure 3. KNN predictions for different k (Q10).

- **k = 1** copies the single nearest country (Israel, 7.2). The prediction depends on one possibly unusual point, and the model fits the training data perfectly ($R^2 = 1.00$) because every point is its own nearest neighbor. This is a sign of **overfitting** (high variance), not of a good model.
- **k = 3, 5, 10** average more countries, so the predictions stabilize around 6.3–6.4, close to the linear regression result.
- A very large k leads to **underfitting** (high bias): with k = 27 the model always predicts the overall mean of 6.57.

6. Comparison and generalization

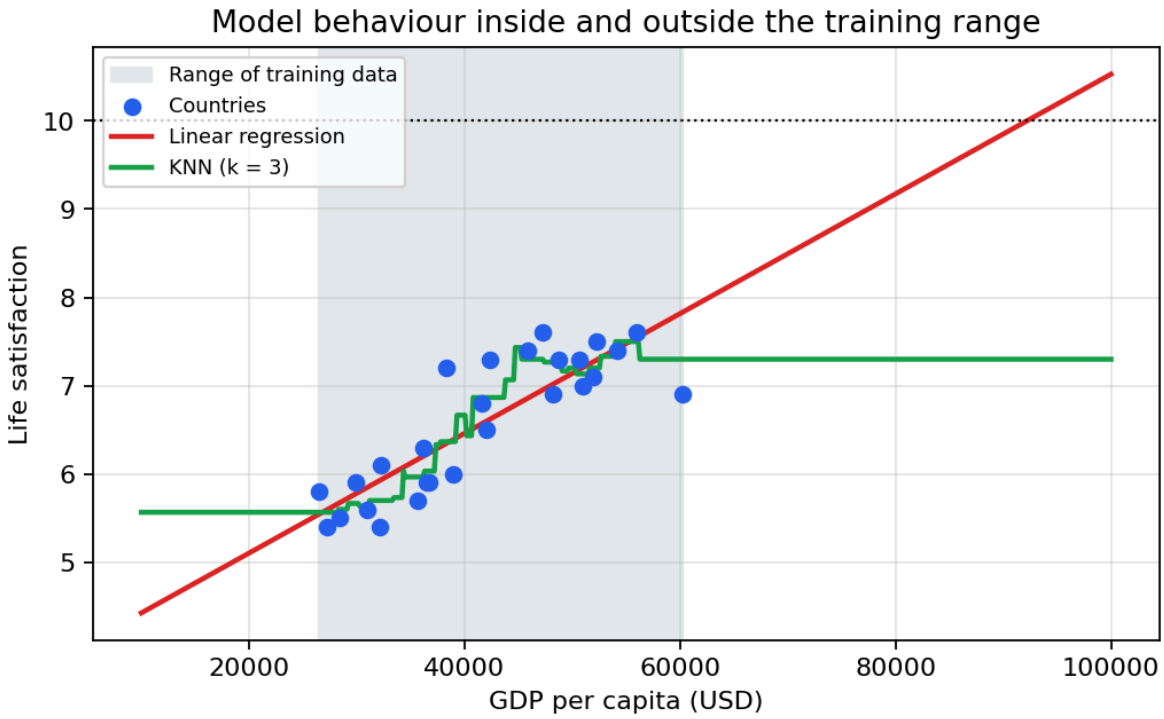


Figure 4. Linear regression and KNN (k = 3) inside and outside the range of the data. The dotted line is the maximum possible score of 10.

	Linear regression	**KNN regression**
Type of learning	model-based: learns two parameters	instance-based: stores all the data
Shape of the prediction	a straight line	a step-like curve that follows local data
Prediction at 37,655.2 USD	6.30	6.33 (k = 3)
Prediction at 100,000 USD	10.53 (above the 0–10 scale)	7.30 (stuck at the richest countries)
Main hyperparameter	none	k, the number of neighbors

Inside the data range both models give similar answers. Outside it they fail in different ways: linear regression keeps growing without limit and soon predicts impossible values, while KNN simply repeats the values of the nearest edge countries. Neither model can be trusted far from the data it was trained on.

Limitations of this analysis

- **No train/test split.** Both models were fitted and evaluated on the same 27 points, so we cannot measure how well they generalize. A held-out test set or cross-validation is needed.
- **Very small sample.** 27 observations make the results sensitive to individual countries.
- **Sampling bias.** Only countries with GDP between 26,000 and 60,000 USD are included.
- **One feature.** Life satisfaction also depends on health, freedom, social support and other factors.
- **Feature scaling.** With several features, KNN would require scaling (for example `StandardScaler`), because it is based on distances.

7. Conclusion

GDP per capita is strongly and positively associated with life satisfaction ($r \approx 0.85$). A simple linear regression explains about 73% of the variance and predicts 6.30 for a country with a GDP of 37,655.2 USD. KNN regression with $k = 3$ gives a very close value of 6.33. The number of neighbors controls the trade-off between overfitting (small k) and underfitting (large k). Because the dataset is small, covers a narrow GDP range and was not split into training and test sets, these results describe the observed countries well but should not be extrapolated to very poor or very rich countries.

Appendix: key code

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.neighbors import KNeighborsRegressor

df = pd.read_csv("lifesat.csv")
X = df[["GDP per capita (USD)"]].values      # shape (27, 1)
y = df[["Life satisfaction"]].values         # shape (27, 1)

lin_reg = LinearRegression().fit(X, y)
print(lin_reg.coef_[0][0], lin_reg.intercept_[0])
print(lin_reg.predict([[37655.2]]))           # ~6.30

for k in [1, 3, 5, 10]:
    knn = KNeighborsRegressor(n_neighbors=k).fit(X, y)
    print(k, knn.predict([[37655.2]]))        # 7.20, 6.33, 6.26, 6.37
```